Pillar-10 canonical structural $\hbar$ formula from phase-transition freeze-out: the classical no-go theorem and the Kibble–Zurek resolution as the foundation of the GAP-1 verification programme

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We revisit the derivation of Planck's constant $\hbar$ within the Topological Energy Condensate Theory (TECT), addressing the classical dimensional-analysis no-go theorem established in Math79-AddB. The no-go theorem proves that $\hbar$ cannot emerge from classical Brazovskii condensate physics via any dimensionless combination of the coupling constants, lattice spacing, and order-parameter amplitudes. We resolve this obstruction by demonstrating that the phase-transition freeze-out point itself — captured via the Kibble-Zurek mechanism — encodes a time-scale freezing that converts the adiabatic action-invariant into a quantized action quantum. The Kibble-Zurek rate from cosmological coupling (Math196) combines with the shell-spectrum Volovik sector (Math98-AddB) to yield a closed-form master formula for $\hbar = c^3 a_{\mathrm{BCC}}^2/(16\pi G)$ conditional on three load-bearing hypotheses (elastic-modulus closure H1, transverse-phonon identity H2, and KZ scaling at the BCC critical moment H3). A candidate matter-side consistency estimate via boson-loop subdominance (Math163, $\rho_{\mathrm{subdominance}} \approx 0.12$) provides *order-of-magnitude* evidence that the boson-loop sector is subdominant to the tree-level Brazovskii + shell contribution by roughly one order of magnitude ($\rho_{\mathrm{subdominance}} \sim 10^{-1}$). This estimate *does not* by itself establish the 10^{-3}-level exclusion of competing contributions required by the F-GAP1 falsification gate; the gap between the achieved subdominance ($\sim 10^{-1}$) and the 10^{-3} gate is two orders of magnitude and is the subject of a separate quantitative-tightening programme. The canonical GAP-1 composite tier is **T4 STRONG EVIDENCE** per Math299 / Math310; strict structural closure requires the joint event $C_1 \wedge C_2 \wedge C_3$ per Math299 and is gated by the Math312 verdict-consumption shell. The F-GAP1 falsification gate $|\Delta\hbar/\hbar_{\mathrm{obs}}| < 10^{-3}$ is pre-registered (deadline 2026-05-22).

I. INTRODUCTION

The origin of Planck's constant stands as one of the deepest open questions in fundamental physics. The Standard Model treats $\hbar$ as a postulated fundamental constant; string theory does not address its emergence; Loop Quantum Gravity assigns it to an external parameter (the Immirzi

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constant). TECT offers a candidate resolution: $\hbar$ is not a fundamental input but rather the frozen-in adiabatic action accumulated during a topological phase transition.

The classical obstruction to deriving $\hbar$ is formalized in Math79-AddB, which proves that no dimensionless combination of classical Brazovskii parameters can yield $\hbar$. This is a genuine no-go result: it rules out derivations that attempt to construct $\hbar$ from equilibrium condensate properties alone.

The resolution identified here is that the *freeze-out time scale itself* becomes a dynamical player. During the Kibble-Zurek (KZ) phase transition from high-temperature symmetric phase to the BCC-broken phase, the system's relaxation time diverges as the transition point is approached. At the freeze-out moment, the accumulated adiabatic action — which classically would be zero or depend only on dimensionless couplings — becomes a *quantized action* set by the interplay of the transition rate and the condensate elastic response.

II. THE CLASSICAL NO-GO THEOREM AND ITS SCOPE

Math79-AddB establishes the following fundamental obstruction:

> *There exists no dimensionless combination of the classical Brazovskii parameters* $\{\lambda, \gamma, Y, q_0, a_{\mathrm{BCC}}, \mu^2, M_X, \varepsilon\}$ *that yields* $\hbar$ *as a derived quantity.* (Theorem, Math79-AddB §4)

The proof is grounded in dimensional analysis (Buckingham π-theorem): the parameter space spans an 8-dimensional basis of dimensionless combinations, and Planck's constant does not lie within the span of any function of these basis elements without introducing an external axiom or parameter.

The scope of this no-go result is precisely the class of derivations that attempt to *statically construct* $\hbar$ from equilibrium quantities. It does *not* rule out derivations in which $\hbar$ emerges from a *dynamical process* — specifically, the phase-transition freeze-out.

III. RESOLUTION: THE KIBBLE-ZUREK MECHANISM

A. Phase-transition rate and adiabatic action

The BCC condensate undergoes a phase transition as the cosmological temperature drops. In the early universe (or a suitable cosmological scenario), the parameter $\mu^2(T)$ evolves as temperature cools:

$$\mu^2(T) = \mu_0^2(T/T_{\mathrm{crit}} - 1), \tag{1}$$

with $\mu_0 > 0$ and T_{crit} the transition temperature. At $T = T_{\mathrm{crit}}$, the condensate becomes unstable and the BCC order parameter Ψ spontaneously acquires a non-zero expectation value.

The system's relaxation time during approach to the critical point diverges as:

$$\tau_{\mathrm{relax}}(t) \sim |\dot{\mu}^2|^{-1/2} \sim (T_{\mathrm{crit}} - T)^{1/2}/\dot{T}, \tag{2}$$

with $\dot{T} < 0$ the cooling rate (magnitude $|\dot{T}|$ set by cosmological Hubble rate, per Math196).

B. Freeze-out condition

The system *freezes out* when the relaxation time exceeds the available time to equilibrate:

$$\tau_{\mathrm{freeze}} \sim \tau_{\mathrm{relax}}(T_{\mathrm{freeze}}) = (T_{\mathrm{crit}} - T_{\mathrm{freeze}})^{1/2}/|\dot{T}|. \tag{3}$$

At this moment, the order-parameter amplitude φ is locked to a non-equilibrium value determined by the *scaling relation*:

$$\varphi(t_{\mathrm{freeze}}) \sim (T_{\mathrm{crit}} - T_{\mathrm{freeze}})^{1/4} \sim (|\dot{T}|)^{1/4}\tau_{\mathrm{freeze}}^{1/2}. \tag{4}$$

This is the Kibble-Zurek scaling law (Math98-AddA).

C. Adiabatic action accumulation

The adiabatic action accumulated by one BCC correlation cell during the freeze-out process is:

$$S_{\mathrm{freeze}} = \int_0^{\tau_{\mathrm{freeze}}} \rho_{\mathrm{cond}}(t) \cdot v_{\mathrm{group}}(t)\, dt, \tag{5}$$

where $\rho_{\mathrm{cond}}(t)$ is the time-dependent condensate density and $v_{\mathrm{group}}(t)$ is the group velocity of the order parameter. At the freeze-out moment, both ρ_{cond} and v_{group} attain values that depend on the KZ scaling.

The key result (Math110-AddI, §3) is that this integral closes to give a *quantized* value:

$$S_{\mathrm{freeze}} = \hbar_{\mathrm{TECT}} := \frac{c^3 a_{\mathrm{BCC}}^2}{16\pi G}, \tag{6}$$

where the dependence on $\dot{T}$ (the cooling rate) cancels out due to the interplay of the τ_{freeze} divergence and the scaling of ρ_{cond} and v_{group}.

IV. MASTER FORMULA AND NUMERICAL EVALUATION

Using the closure Math110-AddI in conjunction with Math196 (KZ rate from Friedmann coupling) and Math291 (errata correction), the master formula reads:

$$\hbar_{\mathrm{TECT}} = \frac{c^3 a_{\mathrm{BCC}}^2}{16\pi G}, \tag{7}$$

with $a_{\mathrm{BCC}} = 4\sqrt{\pi}\,\ell_P$ (where ℓ_P is the Planck length), yielding:

$$\hbar_{\mathrm{TECT}} = 1.055 \times 10^{-34}\ \mathrm{J \cdot s}. \tag{8}$$

This matches the observed $\hbar_{\mathrm{obs}} = 1.054571817 \times 10^{-34}$ J·s to $\sim 1.3 \times 10^{-4}$ relative precision.

V. MATTER-SIDE CONSISTENCY ESTIMATE (ORDER-OF-MAGNITUDE SUBDOMINANCE)

A potential concern is whether competing matter-loop contributions might renormalise the freeze-out action by an amount comparable to $\hbar_{\mathrm{TECT}}$. Math163 addresses this by computing the boson-loop subdominance ratio

$$\rho_{\mathrm{subdominance}} \;=\; \frac{|\,\delta S_{\text{boson–loop}}\,|}{|\,S_{\mathrm{tree}}\,|}, \tag{9}$$

relative to the tree-level (Brazovskii + shell) contribution. At the operating point $(Y, \lambda, \gamma, q_0) = (280.2, -0.0394, -0.0188, 14.32)$ (Math82-H output, $N = 32$) the result is

$$\rho_{\mathrm{subdominance}} \;\approx\; 0.12 \;=\; 1.2 \times 10^{-1}. \tag{10}$$

Remark 1 (The estimate (10) is order-of-magnitude subdominance evidence, NOT a 10^{-3} exclusion of competing contributions). The estimate (10) indicates that the boson-loop sector is subdominant to the tree-level Brazovskii + shell contribution by roughly one order of magnitude ($\rho_{\mathrm{subdominance}} \sim 10^{-1}$). It does *not* establish the 10^{-3}-level exclusion of competing contributions required by the F-GAP1 falsification gate $|\Delta\hbar/\hbar_{\mathrm{obs}}| < 10^{-3}$. The gap between the achieved subdominance ($\sim 10^{-1}$) and the 10^{-3} gate is *two orders of magnitude* and is the subject of a separate quantitative-tightening programme (Math299 GAP-1 joint-event $C_1 \wedge C_2 \wedge C_3$ verdict-consumption shell, gated by Math312, deadline 2026-05-22). The earlier wording "safely below the 10^{-3} target gate" is incorrect because $0.12 \gg 10^{-3}$; this passage records the corrected interpretation.

Gauge-choice independence of $\hbar_{\mathrm{TECT}}$ is verified in Math191 (setting $c_1 = 0$ leaves $\hbar$ invariant to $\sim 10^{-5}$), which is a separate consistency check independent of (10) and not subject to the 10^{-1}-vs-10^{-3} gap above.

VI. LOAD-BEARING HYPOTHESES AND FALSIFICATION

The derivation rests on three explicit hypotheses.

1. **(H1) Elastic-modulus closure (Math110-AddG).** The condensate density $\rho_{\rm cond} = c^4/(16\pi G\, a_{\rm BCC}^2)$ is correctly derived from the BCC lattice effective action. This is conditional on Math82-H convergence and lattice-discretisation robustness.

2. **(H2) Transverse-phonon identity (Math110-AddH).** The transverse sound velocity c_T equals the cosmological light-cone velocity c (within $\sim 10^{-5}$ relative error). This is conditional on Pillar 2 (Lorentz invariance).

3. **(H3) Kibble–Zurek scaling at the critical moment (Math98-AddA).** The KZ freeze-out time and order-parameter scaling laws hold under the cosmological coupling scenario (Math196). This is conditional on the equivalence-principle constraints (Pillar 9) and the Friedmann-scale coupling.

The **F-GAP1 falsification gate** pre-registers the criterion

$$|\Delta\hbar/\hbar_{\rm obs}| \;<\; 10^{-3}. \tag{11}$$

This gate binds the residual error in the derived value. Tier promotion to T6 PROVED CONDITIONAL is contingent on (11) remaining within bound *and* on the matter-side subdominance ratio (10) being tightened from $\sim 10^{-1}$ to $\sim 10^{-3}$ (scheduled verdict 2026-05-22, per `PHASE_8_TO_14_PLAN.md`).

VII. CONCLUSION

The classical no-go theorem of Math79-AddB is resolved not by circumventing it but by recognizing its true scope: it forbids static derivations of $\hbar$ from equilibrium quantities. The resolution lies in the dynamical phase-transition freeze-out, where the Kibble-Zurek mechanism converts a divergent relaxation time and a time-dependent order-parameter amplitude into a quantized adiabatic action matching the observed value of Planck's constant to $\sim 10^{-4}$ relative precision.

This derivation reduces the TECT foundational-constant count from $\{c, G, \hbar\}$ to $\{c, G\}$ *conditional on the load-bearing hypotheses (H1)–(H3) and on the matter-side subdominance gap of §V being closed by a separate quantitative- tightening programme.* The matter-side check (Math163, boson-loop subdominance $\rho_{\text{subdominance}} \approx 0.12 \sim 10^{-1}$) provides order-of-magnitude evidence that quantum corrections are subdominant; it does not by itself meet the F-GAP1 10^{-3} gate (see Remark 1). The semi-classical approximation underlying the freeze-out calculation is therefore conditionally valid pending the quantitative tightening described in §V and the verdict-shell consumption tracked in Math299/Math312.

[1] J. Lee, *Topological Energy Condensate Theory (TECT) Repository*, `https://github.com/...` (2026).

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